

Biophoton Field Map: Milestone 2 Checkpoint

Cluster map, field-boundary result, and verification.

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• 1. The field-boundary question (the analytical ask)

Sonoluminescence / acoustic-cavitation seed works land in bibliographic-coupling community **4**, while the UPE core (Cifra / Popp / Van Wijk / Kobayashi) lands in community **0**, **distinct communities**.

- Sonoluminescence seed works (52): community distribution {'4': 34, '1': 10, '2': 1, '6': 2}
- UPE-core seed works: distribution {'0': 36}

This empirically confirms the physics (single-bubble sonoluminescence, cavitation) is an **adjacency**, not part of the biophoton/UPE core, answering the scope question from the map itself rather than by pre-filtering.

• 2. Sub-field communities (bibliographic coupling)

11 communities over 14807 works (2,392,824 coupling edges). Largest communities:

Community 0: 5193 works, 125 seeds, median year 2012 (UPE / biophoton core).

- **Topics:** Biofield Effects and Biophysics, bioluminescence and chemiluminescence research, Photoreceptor and optogenetics research, Electromagnetic Fields and Biological Effects, Neural dynamics and brain function
- **Top authors:** Michael A. Persinger, Jack A. Tuszyński, Michal Cifra, Ignat Ignatov, Agata Scordino, Francesco Musumeci
- **Most-cited work:** Hallmarks of Cancer: The Next Generation (2011.0, 66806 cites)

Community 1: 3016 works, 13 seeds, median year 2014.

- **Topics:** Ultrasound and Cavitation Phenomena, Advanced oxidation water treatment, Ultrasound and Hyperthermia Applications, Nanoplatfroms for cancer theranostics, Innovative Microfluidic and Catalytic Techniques Innovation
- **Top authors:** Muthupandian Ashokkumar, Oualid Hamdaoui, Slimane Merouani, Kyuichi Yasui, Kenneth S. Suslick, Franz Grieser
- **Most-cited work:** Biodiesel from microalgae (2007.0, 9282 cites)

Community 2: 2869 works, 1 seeds, median year 2015.

- **Topics:** Ultrasound and Cavitation Phenomena, Ultrasound and Hyperthermia Applications, Cavitation Phenomena in Pumps, Minerals Flotation and Separation Techniques, Fluid Dynamics and Mixing

- **Top authors:** Michel Versluis, Claus-Dieter Ohl, Detlef Lohse, Werner Lauterborn, Nico de Jong, Yuning Zhang
- **Most-cited work:** Hydrodynamic and Hydromagnetic Stability (1962.0, 10446 cites)

Community 3: 1974 works, 9 seeds, median year 2011.

- **Topics:** Mitochondrial Function and Pathology, Skin Protection and Aging, Redox biology and oxidative stress, Nitric Oxide and Endothelin Effects, Plant Stress Responses and Tolerance
- **Top authors:** Rafael Radí, Helmut Sies, Barry Halliwell, Ohára Augusto, Pavel Pospíšil, Jens J. Thiele
- **Most-cited work:** The Hallmarks of Aging (2013.0, 15043 cites)

Community 4: 1154 works, 34 seeds, median year 2007 (sonoluminescence / cavitation (adjacency)).

- **Topics:** Ultrasound and Cavitation Phenomena, Quantum Electrodynamics and Casimir Effect, Nuclear Physics and Applications, Ultrasound and Hyperthermia Applications, Electrohydrodynamics and Fluid Dynamics
- **Top authors:** Seth Putterman, Ho-Young Kwak, Kyuichi Yasui, Detlef Lohse, Weizhong Chen, Г. Л. Шарипов
- **Most-cited work:** Electrogenenerated Chemiluminescence and Its Biorelated Applications (2008.0, 2113 cites)

Community 5: 452 works, 0 seeds, median year 2011.

- **Topics:** Planarian Biology and Electrostimulation, Ion channel regulation and function, Developmental Biology and Gene Regulation, Plant and Biological Electrophysiology Studies, Cancer Cells and Metastasis
- **Top authors:** Michael Levin, Min Zhao, Dany Spencer Adams, Richard B. Borgens, Colin McCaig, Brian Reid
- **Most-cited work:** Putting tumours in context (2001.0, 2171 cites)

Community 6: 141 works, 2 seeds, median year 2021.

- **Topics:** Minerals Flotation and Separation Techniques, Ultrasound and Cavitation Phenomena, nanoparticles nucleation surface interactions, Industrial Gas Emission Control, Nonlocal and gradient elasticity in micro/nano structures
- **Top authors:** Kyuichi Yasui, Toru Tuziuti, Wataru Kanematsu, Jun Hu, Muidh Alheshibri, Lijuan Zhang
- **Most-cited work:** Principle and applications of microbubble and nanobubble technology for water treatment (2011.0, 1041 cites)

Community 7: 2 works, 0 seeds, median year 1997.

- **Topics:** Spectroscopy and Chemometric Analyses, Water Quality Monitoring and Analysis
- **Top authors:** James M. Gallas, M. Eisner, Xianglei Cheng, Lixia Zhao, Xu Wang, Jin-Ming Lin
- **Most-cited work:** FLUORESCENCE OF MELANIN-DEPENDENCE UPON EXCITATION WAVELENGTH AND CONCENTRATION (1987.0, 105 cites)

Community 8: 2 works, 0 seeds, median year 1998.

- **Topics:** Photosynthetic Processes and Mechanisms, Photoreceptor and optogenetics research
- **Top authors:** Guenter Albrecht-Buehler
- **Most-cited work:** Autofluorescence of Live Purple Bacteria in the Near Infrared (1997.0, 34 cites)

Community 9: 2 works, 0 seeds, median year 2001.

- **Topics:** Microtubule and mitosis dynamics, Electron Spin Resonance Studies
- **Top authors:** Shyam Sundar Maity, Lalita Das, Sanjib Ghosh, Suranjana Guha, Satinder S. Rawat, Amitabha Chattopadhyay
- **Most-cited work:** Tubulin Conformation and Dynamics: A Red Edge Excitation Shift Study (1996.0, 60 cites)

• 3. Co-authorship structure

- 39,312 authors, 154,448 co-authorship edges, 5,918 research-group communities (labs/collaborations).

• 4. §10 verification

- **Seed coverage:** 245/263 seeds resolved (target ≥ 240 : PASS).
- **Boundary sanity:** sonoluminescence separates from UPE core: PASS.
- **Known-author spot checks** (must appear with sane institution + high seed-connectedness):

AUTHOR	SEED WORKS	INSTITUTION	COUNTRY	ORCID	TOTAL WORKS	CITED BY
Michal Cifra	15	Czech Academy of Sciences, Insti	CZ	0000-0002-8853-9523	163	2,571
Masaki Kobayashi	10	NTT Basic Research Laboratories	JP	0000-0002-9968-1410	252	4,226
Roeland Van Wijk	9				1	31
Eduard van Wijk	7	Laguna Research	US	0000-0002-4432-3042	53	676
Felix Scholkmann	6	University of Bern	CH	0000-0002-1748-4852	253	10,695
Eduard P.A. Van Wijk	5				40	1,150
Vahid Salari	5	University of Calgary	CA	0000-0001-7908-0696	116	1,291
Francesco Musumeci	5	Politecnico di Milano	IT	0000-0002-5554-3366	165	1,815
F. A. Popp	4				105	2,053

Fritz-Albert Popp	2				38	1,033
Eduard P A Van Wijk	1				1	10
M. Kobayashi	1	Tohoku University	JP	0009-0008-3148-143X	54	514

• 5. Deliverables so far

- data/db/fieldmap.sqlite: queryable DB (works, authors, institutions, work_authors, topics, citation/coauthor edges).
- data/exports/*.parquet|csv: all tables.
- data/exports/*.graphml: coauthorship, coupling, cocitation, topic graphs (open in Gephi).
- data/exports/work_communities.csv: per-work sub-field community assignment.
- run_log.md: every prune/cap count.